Subsequences, Universality, and the Arch Factorization

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Abstract

This report covers sequences and subsequences and arch factorization. The connection between arches and universality. It also explores the comparison of two words over finite alphabets using length of k. The words osso and osos have the same subsequences up to the length of k equals 2 but differ at length of k equals 3 because having different subsequences at this length.

1 Introduction

This report will go over the topic of subsequences, universality and arch factorization. each topic will build on top of the previous theory. Sequences of symbols over a finite set make up words within the finite set. This allows for the formation of subsequences which help preserve order within a sequence while allowing symbols to be skipped. These concepts will allow the theory of arch factorization to be explained in this report. This will deepen how words are compared.

2 Preliminaries

Rich introduces the theory of building and analyzing problems through using strings. Rich proposes the question "Given some string s and some language L, is s in L?" [3, Ch. 2] This question serves as the starting point into how we can understand strings within languages and how they interact with each other. Using strings and an alphabet which is described as a finite set denoted by Σ. The members of the alphabet are described as symbols or characters [3, Sec. 2.1]. This concept acts as a building block moving forward in how to set up problems and understand why strings help us answer automata problems.

The aspect of Rich's introduction to strings and finite sets is that the shortest string over an alphabet is the empty string (ε). In 2.1.2 Rich deepens the understanding of what strings can do and introduces the concept of how the length of the string is written and used. The amount of symbols a string has is defined by the length shown as $|s|$. The concept of concatenation which is described as the connection of symbols in strings end to end. Rich uses several examples showcasing how by adding two strings together or additional mathematical equations strings can be used to solve problems. when reading left to right the symbols in the word can be deleted and keeping the symbols remaining in order. [1] Using these core principles of strings and its properties the next step in investigating these principles can be found in other similar works of academia introducing more ways to utilize strings.

3 Universality and the Arch Factorization

My alphabet $\Sigma = \{H, S, O\}$ contains H, S, and O which are used to form words. $\iota(w)$ is the largest k in which you can find every length of k word over a chosen alphabets subsequence of w. Building

on section 2 the introduction of more in depth aspects of strings introduce the concept of arches within a string. In a string to find the arch you must follow the first contiguous block left to right once each symbol appears at least one time the arch is closed and the next arch begins. When you reach the end of the word symbols outside of the last close arch become the rest while the symbols inside the arches identify with that arch. Arches also do not follow a uniform length. One arch might be 5 symbols long while another is only 3. [1]

When focusing on Figure 1 an eight symbol word is cut in two complete arches and nothing is left over. A two symbol pattern that appears in the word is `HS` and appears at symbol 2 and 3. When looking at a 3 symbol pattern `OSH` the pattern has `O` at symbol 5 and `S` at symbol 8. Since the last symbol is symbol 8 `OSH` is not a subsequence of w. Which is why $\iota(w) = 2$ the word contains every two symbol pattern but not every three symbol one.

Every arch is guaranteed to contain every symbol at least once. Take the pattern's first letter from arch 1 and its second letter from arch 2. Since the positioning of arch 1 is entirely before arch 2 the positions are in order. Therefore the pattern is a subsequence of the word. When a word has k arches take a letter in the pattern from each arch. The word will contain every pattern of length k. [1]

Position:	1	2	3	4	5	6	7	8
$w =$	S	H	S	S	O	O	H	S
	Arch 1					Arch 2		

Figure 1: Arch factorization of $w =$ `SHSSOOHS` over $\Sigma = \{H, S, O\}$: `SHSSO` | `OHS`, with rest ε and universality index $\iota(w) = 2$.

4 Why Equivalence

Continuing delving into length k thus focusing on the equality between two sets. Another example by comparing the words `osso` and `osos` by using length of k one can come to the solution that length of $k = 2$ as all possible $k = 2$ pairs match. When going to $k = 3$ not all possible combinations are proven true so length of k at 3 fails and stays at 2. Two words u and v when compared using length of k are equivalent up to the k length. This means every sub sequence up to length of k will match between the two words. [4, 2] The words are equivalent up to length of k equals 2 so `osso` $\sim_2$ `osos`. Although sso does occur as a subsequence of osso it does not appear in the word osos. So `osso` $\not\sim_3$ `osos` Table 1 shows the possible patterns of length of k over three symbols.

i	1	2	3	4
u_i	o	s	s	o
v_i	o	s	o	s

$$u_2u_3u_4 = \texttt{sso}, \qquad \nexists 1 \le i < j < \ell \le 4 : v_iv_jv_\ell = \texttt{sso}.$$

$$\texttt{osso} \sim_2 \texttt{osos}, \qquad \texttt{osso} \not\sim_3 \texttt{osos}.$$

Figure 2: Showing Osso and Osos and their possible patterns over length of k

Table 1: Number of candidate patterns of exactly length k over a three-symbol alphabet: $|\Sigma|^k = 3^k$. These are candidate words, not counts of subsequences present in a particular word.

k	3^k
5	243
10	59,049
20	3,486,784,401
30	205,891,132,094,649

5 Concluding Remarks

The report explored subsequences and its relations to strings and finite sets. Introducing the concepts of arch factorization added another layer to the understanding of subsequences and their part in relation to how strings and finite sets interact. By focusing on arches and how they allow the factorization of words allowed for the understanding of k lengths. By comparing osso and osos the witness sso was not a matching subsequence at length of k equals 3 showcases how length of k is measured.

References

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[2] Paweł Gawrychowski, Maria Kosche, Tore Koß, Florin Manea, and Stefan Siemer. Efficiently testing Simon's congruence. In *38th International Symposium on Theoretical Aspects of Computer Science (STACS 2021)*, volume 187 of *LIPIcs*, pages 34:1–34:18. Schloss Dagstuhl, 2021.

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